

Emergency Department Performance Review

Hospital comparisons across Alabama, Louisiana, and Mississippi

Project Overview

Hospital leadership needs a clear way to identify emergency department performance gaps and ` further operational review. This project compared hospital-reported ED duration and the percentage of patients who left before being seen with matching state benchmarks.

I used Python to prepare the data, SQL in BigQuery to join datasets and analyze differences, Google Sheets to validate a summary through a pivot table and chart, and Tableau to communicate findings through an interactive dashboard.

Business Questions

1. What percentage of hospitals had comparable scores for each measure?
2. Which hospitals reported the largest differences above their state benchmark?
3. How did benchmark comparisons vary by state, hospital type, ownership, and county/parish?
4. What actions could leadership take based on the findings?

Data Sources

The analysis used three public CMS datasets:

- ★ Hospital General Information
<https://data.cms.gov/provider-data/dataset/xubh-q36u?utm>
Download date: September 30, 2026
Release/Update date: August 13, 2026
- ★ Timely and Effective Care - Hospital
<https://data.cms.gov/provider-data/dataset/yv7e-xc69?utm>
Download date: September 30, 2026
Release/Update date: August 13, 2026
- ★ Timely and Effective Care - State
<https://data.cms.gov/provider-data/dataset/apyc-v239?utm>

Download date: September 30, 2026
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The geographic scope included Alabama, Louisiana, and Mississippi.

Two measures were analyzed separately:

- **OP_18b – ED duration:** October 1, 2024–September 30, 2025; scores measured in minutes.
- **OP_22 – Patients who left before being seen:** January 1–December 31, 2024; scores measured as percentages.

The prepared hospital score dataset contained 592 records representing 296 hospitals, with one record per hospital for each selected measure.

Preparation and Quality Checks

I standardized column names, preserved facility identifiers as text, converted scores to numeric values, and parsed reporting dates.

Of the 592 hospital score records, 129 contained “Not Available.” These were retained as missing scores rather than changed to zero. Actual zero scores remained valid numeric values.

I checked join keys for missing values and duplicates and verified reporting dates. The prepared datasets contained:

- 369 hospital general-information records.
- 592 hospital measure records.
- 6 state benchmark records.

The joined dataset retained all 592 hospital measure records. Validation found no unmatched hospital-information records, unmatched state benchmarks, or hospital-information state mismatches.

Analysis Approach

I used LEFT JOINS to connect hospital scores with facility characteristics by facility ID and with state benchmarks by state, measure ID, start date, and end date.

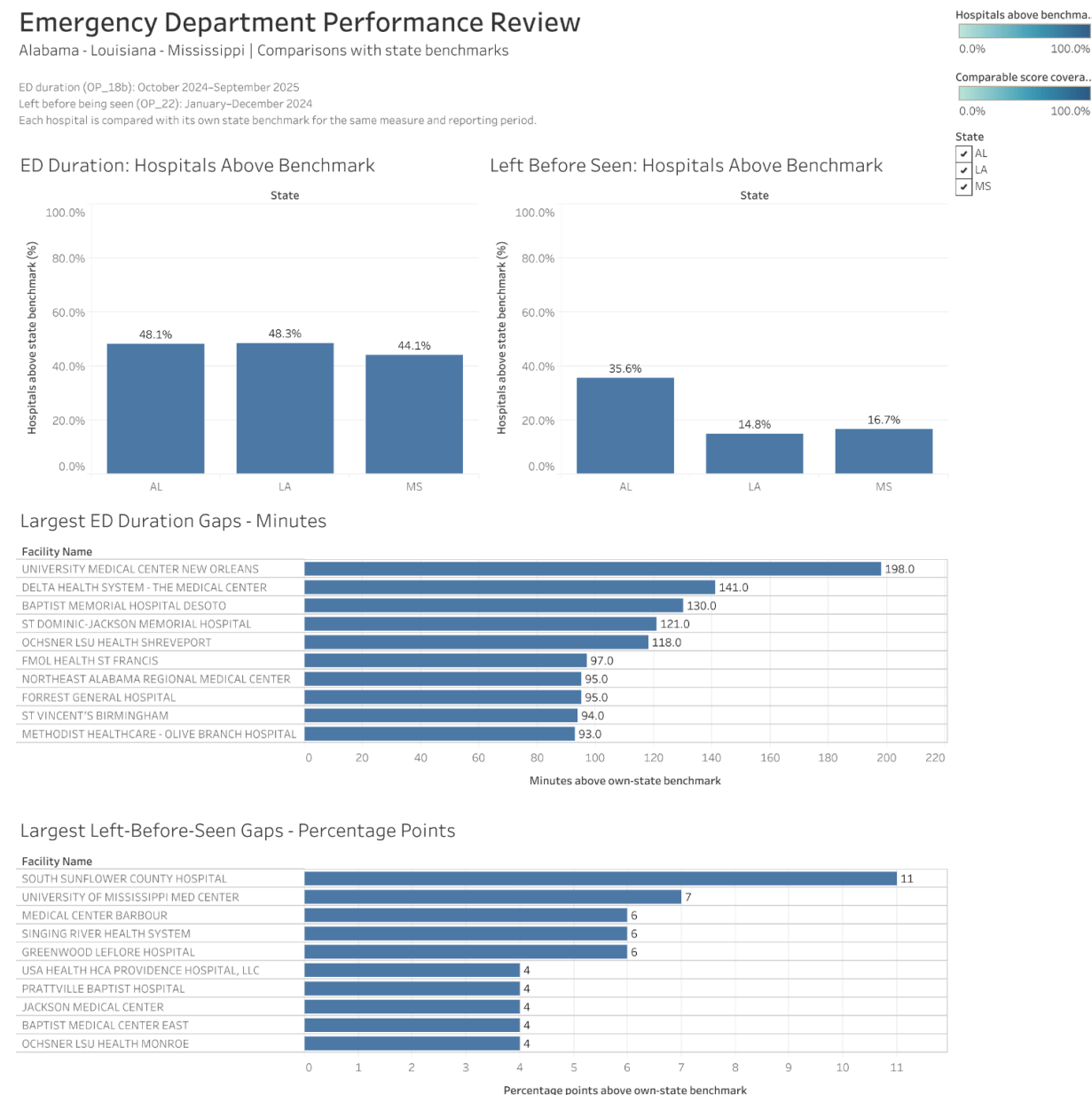
For each comparable record, I calculated:

Difference from state benchmark = Hospital score – State benchmark

Positive differences indicated longer reported ED duration or a higher percentage of patients leaving before being seen than the matching state benchmark.

I analyzed score coverage, state comparisons, hospital rankings, hospital type, ownership, and county/parish patterns. Hospital percentages above benchmark used only records with comparable scores.

I also reproduced a state summary in Google Sheets and built a Tableau dashboard. Dashboard results matched the SQL outputs, and a State filter recalculated the top 10 hospital gaps within the selected state.



Hospitals Above State ED Duration Benchmark by County/Parish

Available Comparisons by State and Measure

State	OP_18b	OP_22
AL		
LA		
MS		

Interactive dashboard showing hospital benchmark comparisons, score coverage, and geographic patterns.
(Link found below)

Key Findings

1. ED duration gaps affected a substantial share of comparable hospitals

For OP_18b, 48.1% of comparable Alabama hospitals, 48.3% of Louisiana hospitals, and 44.1% of Mississippi hospitals reported ED duration above their own state benchmark.

University Medical Center New Orleans had the largest observed gap: 327 minutes compared with Louisiana's 129-minute benchmark, a difference of 198 minutes.

These results identify facilities for additional review. They do not establish the reasons for the differences.

2. Left-before-seen comparisons showed a different pattern

For OP_22, 35.6% of comparable Alabama hospitals exceeded their state benchmark, compared with 14.8% in Louisiana and 16.7% in Mississippi.

South Sunflower County Hospital had the largest observed gap: 14% compared with Mississippi's 3% benchmark, a difference of 11 percentage points.

These percentages describe hospitals exceeding their own state benchmark. They are not statewide patient departure rates.

3. Missing scores limited comparison coverage

Comparable score coverage varied by state and measure:

- Alabama: 90.8% for ED duration and 83.9% for left before being seen.
- Louisiana: 79.5% and 78.6%, respectively.
- Mississippi: 70.1% and 68.0%, respectively.

Missing scores excluded some hospitals from performance comparisons. Coverage therefore needs to be considered alongside the findings.

Recommendations

- **Prioritize facility-level operational review**

Begin with hospitals showing the largest positive benchmark gaps. Review internal patient-flow timestamps, visit volumes, staffing patterns, and process delays to investigate possible explanations.

➤ **Investigate left-before-seen patterns**

At hospitals with elevated OP_22 scores, review departure patterns by time of day, arrival volume, and time to initial assessment. Use this additional evidence to select and evaluate potential workflow changes.

➤ **Review unavailable scores**

Examine reporting footnotes and measure eligibility before treating unavailable scores as reporting problems. Distinguish unavailable comparisons from numeric zero performance.

➤ **Evaluate changes using consistent measures**

If leadership implements changes, track the same measure definitions over comparable periods and account for differences in patient mix and visit volume. This project provides a basis for prioritizing review; it does not measure the impact of an intervention.

Limitations

The analysis is observational and does not establish causation or overall hospital quality.

Hospitals were compared with their own state benchmarks, which differ across states. Above-benchmark percentages do not constitute a direct ranking of statewide performance.

The two measures cover different reporting periods and were not combined into a single performance score.

Unavailable scores may affect the representativeness of comparisons. Small hospital-type, ownership, and county groups require cautious interpretation.

County/parish results describe hospitals located in those areas, rather than outcomes for all residents. Some locations, including East Baton Rouge Parish, did not display on the Tableau map because generated coordinates were missing. Their records remained included in the SQL analysis.

Deliverables

- Python notebook documenting data preparation and validation.
- BigQuery SQL queries and exported analysis results.
- Google Sheets analysis tracker, pivot table, and chart.
- Tableau dashboard with state comparisons, hospital gaps, geographic patterns, score coverage, and a State filter.
- Written case study documenting findings, recommendations, and limitations.

Portfolio Links

Interactive Tableau Dashboard:

https://public.tableau.com/views/HealthcareEDAnalysisProject/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

GitHub Repository: